

# Technical Document #659: Data Engineering - Comprehensive Overview

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Schema evolution handles changes in data structure over time. Schema registries manage schema versions and ensure backward compatibility.

Data lineage tracks the origin and movement of data through systems. It is essential for debugging, compliance, and impact analysis.

Partitioning divides data into smaller, more manageable pieces. It improves query performance by reducing the amount of data scanned.

ETL (Extract, Transform, Load) is a process that extracts data from source systems, transforms it to fit business rules, and loads it into a target data store.

Artificial intelligence continues to reshape industries from healthcare to finance. Machine learning models can now perform tasks that were once thought to require human intelligence.

Data-driven decision making has become essential for modern businesses. Organizations that leverage data effectively gain a significant competitive advantage.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

## Key Takeaways:

- Data lineage tracks the origin and movement of data through systems.
- Data pipelines automate the movement and transformation of data between systems.
- Data warehouses store structured data optimized for analytical queries.